

Ultrathin Yet Effective: 90 nm ZnF₂ Layer for Stabilizing Zinc–Metal Anodes

Viet Phuong Nguyen,^{*} Minji Park, Young-Woon Byeon, Suim Lim, Kanghoon Yim, Minsub Oh, Seungmin Hyun, Eun-chae Jeon,^{*} and Seung-Mo Lee^{*}



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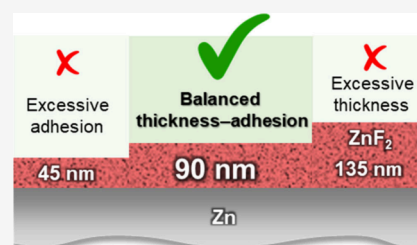


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Supporting Information

ABSTRACT: The practical use of aqueous zinc-ion batteries is limited by dendrite formation and interfacial degradation at the Zn-metal anode. Here, we demonstrate that an ultrathin ZnF₂ interfacial coating, merely 90 nm thick, significantly enhances anode stability by suppressing side reactions, promoting uniform Zn deposition, and providing moderate mechanical adhesion. Symmetric cells with ZnF₂-coated Zn achieved lifespans of 1500 h at 0.5 mA cm⁻² and 500 h at 3.0 mA cm⁻². Full cells with Mn_xV₂O₅ cathodes retained 82% capacity after 2000 cycles. Crucially, nanoscratch tests revealed that the optimum thickness of ZnF₂ films provided reasonable interfacial toughness, offering new insights into the mechanical–electrochemical codesign of artificial protective layers, factors that have often been overlooked or insufficiently investigated so far. This study advances surface engineering for Zn anodes and introduces interfacial mechanics as a design parameter for durable artificial protective layers in aqueous battery systems.



Metallic zinc (Zn) is considered a promising anode material for aqueous zinc-ion batteries (ZIBs) due to its low cost, environmental benignity, and high theoretical capacity (5854 mAh cm⁻³ or 820 mAh g⁻¹).^{1–3} However, its practical application is severely hindered by persistent challenges such as uncontrolled dendritic growth and parasitic side reactions, including corrosion and hydrogen evolution in aqueous electrolytes.^{4–7} These issues result in poor reversibility and low Zn utilization, ultimately compromising battery performance and cycle life.

To overcome these limitations, various strategies have been explored, including anode structural design,⁸ separator modification,⁹ electrolyte optimization,³ and artificial interphase engineering.¹⁰ Among them, the construction of a stable solid–electrolyte interphase (SEI) has emerged as one of the most promising approaches.^{11–14} A well-designed SEI can guide uniform Zn²⁺ deposition and mitigate direct contact between the Zn anode and the aqueous electrolyte, thereby suppressing dendrite formation and parasitic reactions.^{11–14}

Inspired by the success of LiF-rich SEIs in lithium–metal batteries, ZnF₂-rich SEIs have attracted growing interest for ZIBs.^{11–14} ZnF₂ is known for exhibiting high Zn²⁺ ionic conductivity, excellent mechanical hardness, and strong chemical/electrochemical stability, attributes that make it a highly desirable interphase material.^{15,16} Common approaches to incorporating ZnF₂ onto Zn surfaces include ex situ coatings^{17,18} and in situ electrolyte decomposition.^{4,10} Existing coating methods are often complex and yield relatively thick ZnF₂ layers (≈9.3 μm¹⁷ or ≈1 μm¹⁹), while in situ SEI

formation is difficult to control.¹⁶ These drawbacks limit adaptability and cannot be able to fully suppress dendrite growth during cycling. Therefore, simply constructing a thin, well-controlled ZnF₂-rich SEI to improve Zn anode stability remains a challenge. The thickness of artificial coatings has a significant impact on the electrochemical performance of Zn-metal anodes by influencing both ionic transport and mechanical properties.²⁰ An ideal coating should balance ionic conductivity and mechanical strength, while enabling rapid and uniform Zn-ion permeation.^{21,22} Therefore, coating thickness and mechanical strength are considered key factors when fabricating artificial SEI layers on Zn-metal anodes. Unfortunately, the effects of ZnF₂-coating thickness have not been thoroughly investigated in previous research.

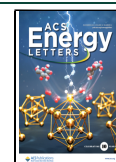
In this study, we demonstrate that a precisely controlled, ultrathin ZnF₂ layer (~90 nm), deposited by atomic layer deposition (ALD), can serve as an effective artificial SEI to stabilize the Zn-metal anode. This nanoscale coating simultaneously suppresses dendritic growth, reduces side reactions, and promotes highly reversible Zn plating/stripping. Symmetric cells with ZnF₂-coated Zn anodes exhibits extended cycling stability over 500 h at 3.0 mA cm⁻² and 3.0 mAh cm⁻²,

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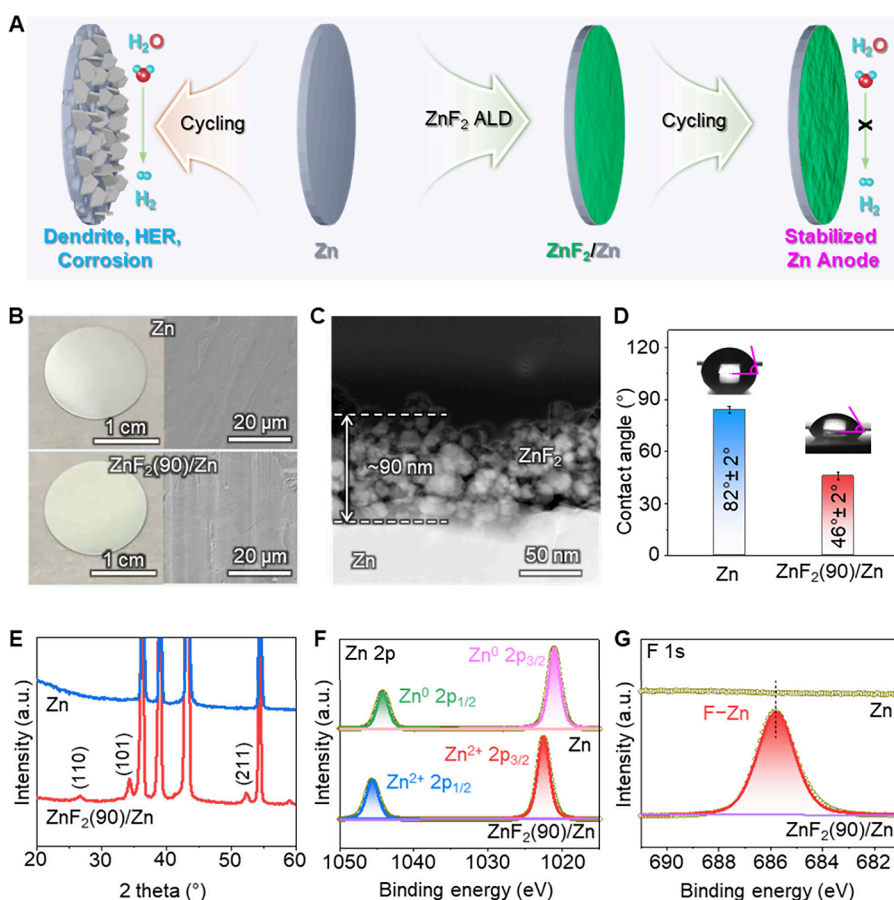


Figure 1. Structural and interfacial characterization of the dendrite-free ZnF₂(90)/Zn anode. (A) Schematic illustration comparing the Zn plating/stripping behavior of bare Zn and ALD-coated ZnF₂(90)/Zn anodes. While bare Zn suffers from dendrite growth and parasitic reactions, the ultrathin 90 nm ZnF₂ layer enables uniform Zn deposition and enhanced stability. (B) Optical and SEM images of bare Zn and ZnF₂(90)/Zn. (C) Cross-sectional STEM-HAADF image showing the thickness of the ZnF₂ layer in ZnF₂(90)/Zn. (D) Contact angle measurements between 1 M ZnSO₄ electrolyte and the Zn or ZnF₂(90)/Zn surfaces, revealing improved wettability after ZnF₂ coating. (E) XRD patterns confirming the presence of ZnF₂. (F, G) High-resolution XPS spectra of Zn 2p (F) and F 1s (G), validating the chemical composition of the ZnF₂ interfacial layer.

while full cells using Mn_xV₂O₅ (MVO) cathodes retain 82% capacity after 2000 cycles.

Crucially, beyond electrochemical testing, we incorporated nanoscratch-based mechanical analysis to investigate the interfacial toughness of the ZnF₂ film. Our findings reveal a strong correlation between film thickness and mechanical adhesion, where excessively thick coatings compromise interfacial integrity. The 90 nm ZnF₂ film achieves an optimal balance between ionic accessibility and mechanical robustness, key criteria for long-term stability. This work not only introduces a scalable and controllable approach to SEI engineering, but also emphasizes critical yet underexplored role of interfacial mechanical properties, which directly impact long-term cycling stability in metal-anode battery systems.

■ PREPARATION AND CHARACTERIZATION OF ZNF₂/ZN

We anticipated that an ultrathin ZnF₂ layer could effectively suppress dendrite growth and side reactions while maintaining fast ion transport for superior rate performance (Figure 1A). To achieve precise thickness control with uniform coverage, we paid attention to ALD, which is a robust technique for conformal thin film deposition in energy storage applications.^{23–26} Utilizing ALD enabled us to modify the Zn foil

surface by a ZnF₂ layer (i.e., ZnF₂/Zn) with impeccable coverage and the desired thickness. To explore the effects of different thicknesses on the electrochemical performance, we deposited ultrathin ZnF₂ layers with various thicknesses by adjusting the number of ALD cycles. Specifically, ZnF₂ thicknesses of 45, 90, and 135 nm were obtained after 315, 625, and 935 cycles, respectively, with the corresponding electrodes denoted as ZnF₂(45)/Zn, ZnF₂(90)/Zn, and ZnF₂(135)/Zn. Preliminary electrochemical screening revealed that ZnF₂(90)/Zn delivered optimal performance, balancing protection and ionic transport, and was therefore selected for comprehensive material characterization.

The morphological changes of the Zn foil following ZnF₂ ALD were investigated using scanning electron microscopy (SEM). As depicted in Figure 1B, the Zn foil exhibited shinier appearance after the ZnF₂ deposition, although its overall morphology remained largely unchanged. A rough and uneven surface of the bare Zn foil, resulting from the initial manufacturing process, was still visible on the ZnF₂(90)/Zn (Figure S1). Cross-sectional scanning transmission electron microscopy with high-angle annular dark-field (STEM-HAADF) imaging confirmed that the ZnF₂ layer on the Zn substrate reaches a thickness of approximately 90 nm after 625 ALD cycles (Figure 1C). In the high-resolution TEM

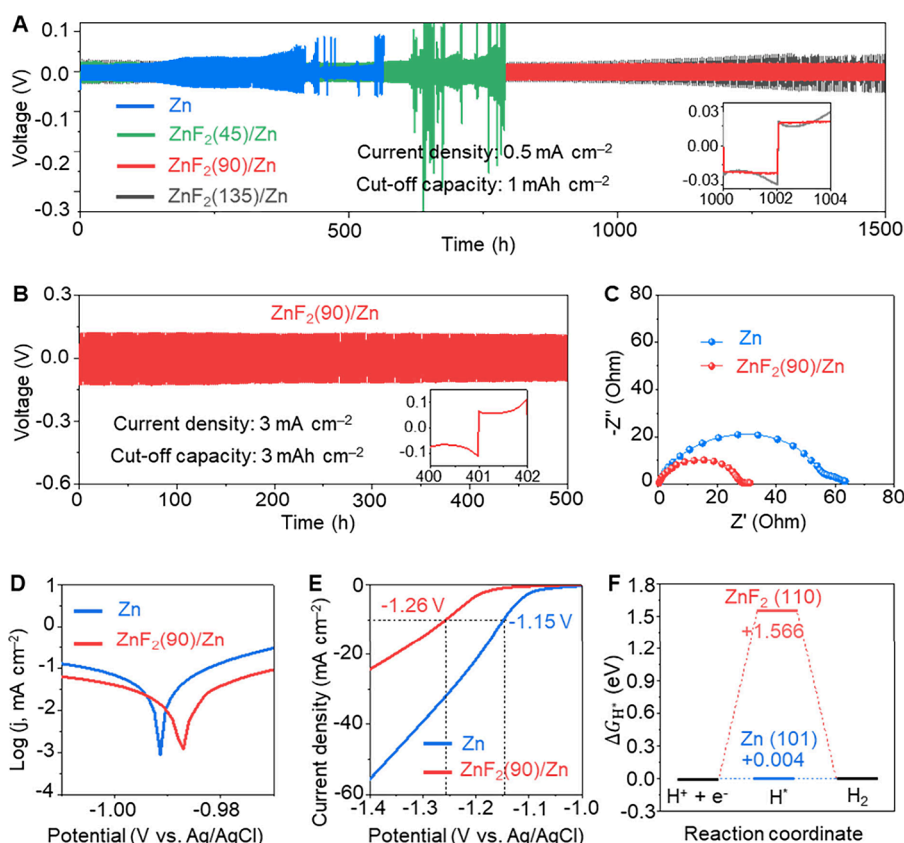


Figure 2. Electrochemical stability and suppression of side reactions in ZnF₂(90)/Zn. (A) Cycling performance of symmetric cells assembled with bare Zn, ZnF₂(45)/Zn, ZnF₂(90)/Zn, and ZnF₂(135)/Zn electrodes at a current density of 0.5 mA cm⁻². The ZnF₂(90)/Zn electrode exhibits superior long-term stability. (B) Cycling performance of the symmetric cell assembled with ZnF₂(90)/Zn electrodes at a high current density of 3.0 mA cm⁻². (C) The EIS spectra of symmetric cells. (D) Tafel plots illustrating the corrosion behavior of Zn and ZnF₂(90)/Zn anodes in aqueous electrolyte. (E) Linear sweep voltammetry curves for evaluating the HER activity of bare Zn and ZnF₂(90)/Zn electrodes. (F) Comparison of the ΔG_H^{*} values for Zn (101) and ZnF₂ (110).

(HRTEM) image, an interplanar distance of 0.24 nm was observed and could be attributed to the (200) plane of ZnF₂ (Figure S2). Energy-dispersive X-ray spectroscopy (EDX) elemental mappings revealed the uniform distribution of Zn and F on the surface (Figure S3). Notably, the contact angle between ZnF₂(90)/Zn and the electrolyte (1 M ZnSO₄ solution) decreased to ~46°, markedly lower than that of bare Zn (~82°) (Figure 1D). This suggested that the ZnF₂ layer enhances the uniform wetting of the electrolyte across the anode surface, thereby enabling the homogeneous distribution of Zn-ion flux at the interface.^{17,27} Moreover, the ionic conductivity of the ZnF₂ layer was roughly calculated to be as high as 1.17×10^{-4} S cm⁻¹ (Figure S4), which indicates its strong ability to facilitate Zn²⁺ transport. The crystal structure and composition of the ZnF₂ layer were further examined via X-ray diffraction (XRD) and X-ray photoelectron spectroscopy (XPS). The emergence of ZnF₂ peaks in the XRD patterns confirmed the presence of crystalline ZnF₂ on the Zn foil (Figure 1E). In the XPS spectrum of ZnF₂(90)/Zn, peaks at 1022.5 and 1045.6 eV correspond to the Zn²⁺ 2p_{1/2} and Zn²⁺ 2p_{3/2}, respectively (Figure 1F).²⁸ Notably, no Zn⁰ signal was detected in the high-resolution Zn spectrum, indicating complete coverage by the ZnF₂ layer. In the F 1s spectrum, the peak centered at 685.8 eV could be assigned to the F–Zn bonds (Figure 1G).²⁹ These results confirmed that the ultrathin ZnF₂ layer was successfully deposited on the surface of Zn foil.

■ ELECTROCHEMICAL STABILITY OF ZNF₂/ZN

The Zn-metal anode achieves stabilization when dendrite growth and side reactions, such as corrosion and HER, are effectively suppressed. The stability of ZnF₂/Zn was first investigated through long-term Zn plating and stripping tests in symmetric cells. We found that, at a current density of 0.5 mA cm⁻² and a capacity of 1 mAh cm⁻², the cell using ZnF₂(90)/Zn was capable of providing an ultralong cycling lifespan of more than 1500 h, maintaining a small voltage hysteresis of ~20 mV (Figure 2A). This performance surpassed that of all other tested configurations. The cell with bare Zn showed a continuous increase in voltage hysteresis, followed by a sudden voltage drop at 350 h. With a 45 nm ZnF₂ coating on the Zn foil, the cycle life of the cell using ZnF₂(45)/Zn extended to about 700 h before failing due to rapid voltage fluctuations. Although the cell using ZnF₂(135)/Zn could work for 1500 h, it consistently exhibited high voltage polarization. This suggested that an excessively thin ZnF₂ layer may be insufficient to suppress dendrite formation, while an overly thick layer could hinder Zn-ion transport, leading to increased voltage hysteresis. The superior performance of ZnF₂(135)/Zn may also be attributed to the adhesion strength between the ZnF₂ layer and the Zn substrate, as discussed in the next section. When the current density and cutoff capacity increased to 2 mA cm⁻² and 2 mAh cm⁻², respectively, the same phenomena were observed (Figure S5). Notably, the cell using ZnF₂(90)/Zn was still

able to operate for more than 500 h at a high current density and cutoff capacity of 3 mA cm^{-2} and 3 mAh cm^{-2} , respectively (Figure 2B). Additionally, the electrochemical impedance spectra (EIS) measurement at the cycle 10th revealed the smaller charge transfer resistance of the cell using $\text{ZnF}_2(90)/\text{Zn}$ compared to that of the cell using Zn (Figure 2C). Density functional theory (DFT) calculations showed that the diffusion barrier of Zn ions on ZnF_2 is only 0.19 eV (Figure S6), which is considerably lower than that on ZnO, another potential protective coating material.³⁰ This low diffusion barrier facilitates the homogeneous distribution of Zn^{2+} ,³¹ thereby promoting uniform Zn deposition. These findings demonstrate that the ZnF_2 layer significantly suppresses dendrite formation, improves electrochemical stability, and enhances the cycling performance of the $\text{ZnF}_2(90)/\text{Zn}$ anode.

To evaluate the ability of the ZnF_2 layer to inhibit side reactions, we performed linear scanning voltammetry tests. As shown in Figure 2D, the corrosion potential of the $\text{ZnF}_2(90)/\text{Zn}$ electrode (-0.98653 V vs Ag/AgCl) was more positive than that of the bare Zn foil electrode (-0.99169 V vs Ag/AgCl). Correspondingly, the ZnF_2 layer reduced the corrosion current from 0.2852 to $0.0922 \text{ mA cm}^{-2}$, suggesting enhanced corrosion resistance. The HER on the $\text{ZnF}_2(90)/\text{Zn}$ electrode was also found to be suppressed. The HER was also effectively suppressed on the $\text{ZnF}_2(90)/\text{Zn}$ electrode, as evidenced by a more negative HER potential and a higher Tafel slope compared to the bare Zn electrode (Figure 2E and Figure S7). DFT calculations revealed that Zn (101) has a hydrogen adsorption free energy (ΔG_{H}^*) value close to zero (Figure 2F). This indicated efficient kinetics of hydrogen intermediates, resulting in high HER activity.³² In contrast, the extremely high ΔG_{H}^* on ZnF_2 (110) makes HER thermodynamically unfavorable. In addition, the poor conductivity of ZnF_2 further hinders electron transfer (Figure S8), thereby suppressing the HER. These results confirm that the ZnF_2 layer provides significant protection against side reactions on the Zn-metal anode.

To gain further insight into the above results, we disassembled the symmetric cells after 375 cycles and conducted further analyses. We observed that the cycled bare Zn anode exhibited severe corrosion and was covered with sharp Zn dendrite flakes (Figure 3A,B). Due to the inherently nonuniform electric field distribution, Zn deposition occurred randomly and further evolved into flaky dendrites oriented in different directions. Over time, these flakes detached and lost electrical contact with the substrate, leading to the formation of “dead” Zn and an increase in charge transfer resistance.³³ In sharp contrast, the cycled $\text{ZnF}_2(90)/\text{Zn}$ anode possessed a smooth surface, free of Zn dendrites (Figures 3C,D and S9). This improvement is attributed to ZnF_2 , which was believed to homogenize Zn-ion flux, suppress the “tip effect,” and promote the parallel deposition of Zn.³⁴ To further investigate the phase evolution, we performed XPS measurements. After cycling, the Zn^{2+} signals were detected on the bare Zn anode, indicating the formation of byproducts, such as $\text{Zn}_4\text{SO}_4(\text{OH})_6 \cdot 5\text{H}_2\text{O}$.¹⁹ Meanwhile, no Zn^0 signal was observed on the surface of the cycled $\text{ZnF}_2(90)/\text{Zn}$ anode, suggesting that the Zn metal remained fully covered by ZnF_2 . Notably, the F[−] signal remained nearly unchanged before and after cycling, demonstrating that the ZnF_2 layer was robust and effectively protected the Zn anode throughout the cycling process. Similar phenomena were also observed with the $\text{ZnF}_2(90)/\text{Zn}$

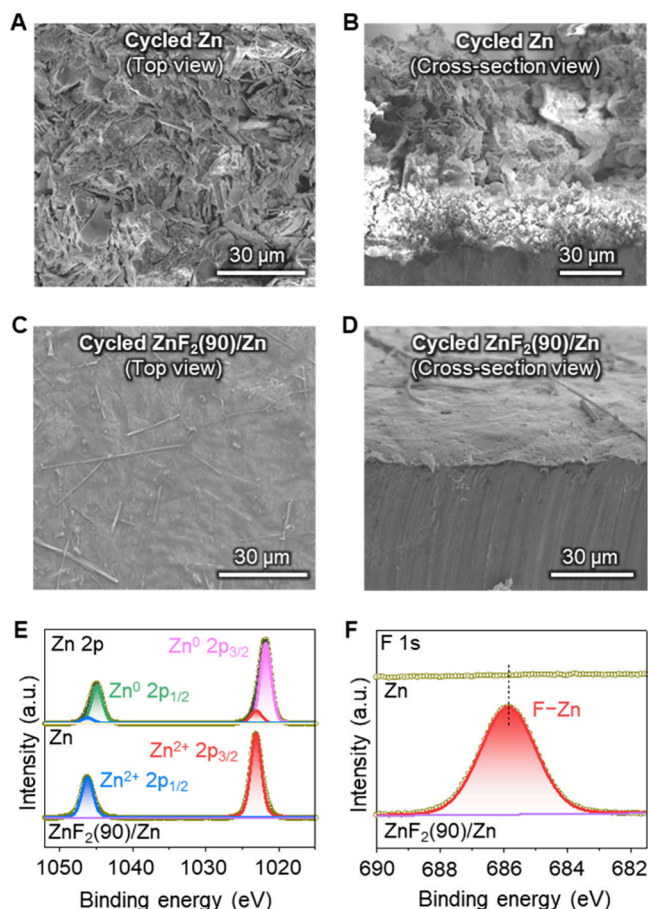


Figure 3. Morphological and chemical characterization of the cycled Zn and $\text{ZnF}_2(90)/\text{Zn}$ electrodes. (A, B) Top-view (A) and cross-sectional (B) SEM images of the Zn electrode after cycling, showing extensive surface roughening and dendritic growth. (C, D) Top-view (C) and cross-sectional (D) SEM images of the $\text{ZnF}_2(90)/\text{Zn}$ electrode after cycling, revealing a smooth and compact surface morphology. (E, F) High-resolution XPS spectra of Zn 2p (E) and F 1s (F) for cycled Zn and $\text{ZnF}_2(90)/\text{Zn}$ electrodes, confirming the chemical stability of the ZnF_2 layer and the suppression of degradation byproducts.

electrode in the symmetric cell after cycling at a higher current density (Figure S10), validating the long-term stability and integrity of the ZnF_2 layer.

INTERFACIAL MECHANICAL CHARACTERISTICS OF ZnF_2 FILM

To gain deeper insight into the origin of the exceptional cycling stability imparted by the ultrathin ZnF_2 interfacial layer, we examined its interfacial mechanical characteristics via nanoscratch tests. We attempted nanoscratch tests of ZnF_2 films directly deposited on Zn foil. However, the inherently high surface roughness and pronounced morphological irregularities of commercial Zn foils resulted in inconsistent penetration depths and irreproducible data, which made reliable analysis nearly impossible. For this reason, we adopted flat Si wafers as a practical alternative. Their smooth surface provided a stable platform that enables reproducible nanoscratch testing and systematic evaluation of thickness-dependent adhesion trends. Although the substrate mismatch may influence absolute values of crack initiation or delamination resistance, it is believed that the comparative results still offer

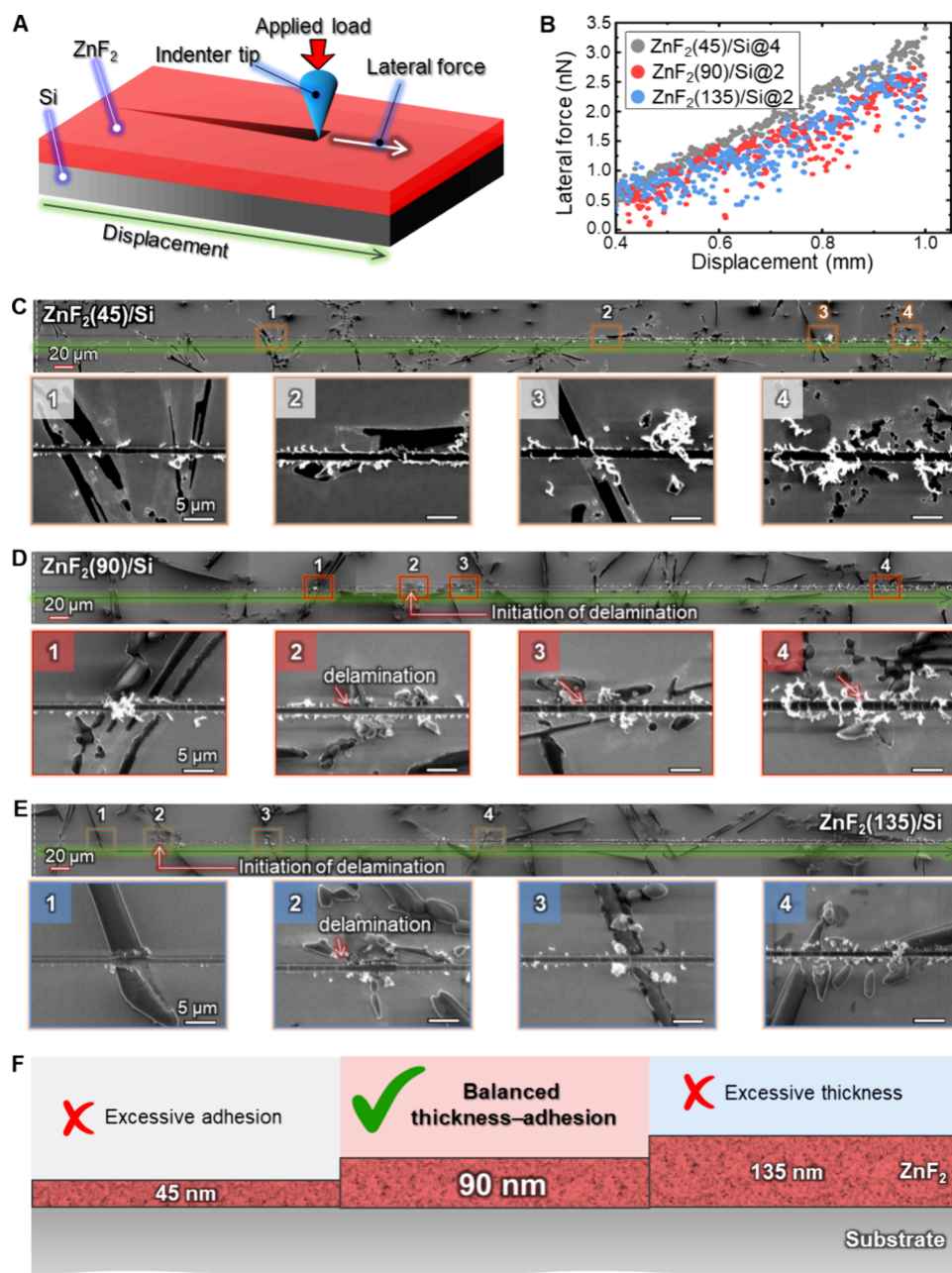


Figure 4. Interfacial mechanical characteristics of ZnF₂ thin films deposited on Si wafer, examined via nanoscratch testing. (A) Schematic illustration of the nanoscratch test, where a progressive normal load was applied to the film surface while recording the resulting lateral (frictional) force during lateral displacement. (B) Lateral force versus displacement curves for ZnF₂ films deposited on Si wafer. (C–E) SEM images of ZnF₂(45)/Si, ZnF₂(90)/Si, and ZnF₂(135)/Si, respectively, after scratching. The results showed minimal film damage or no delamination, demonstrating excellent mechanical integrity. In contrast, ZnF₂(90)/Si and ZnF₂(135)/Si revealed early onset and propagation of delamination during scratching, indicating reduced adhesion with increasing film thickness. These results collectively suggested that thinner ZnF₂ layers exhibit more robust interfacial adhesion, while thicker coatings are more susceptible to mechanical failure under applied stress. (F) Presumable balance between thickness and adhesion strength.

meaningful insights to elucidate general thickness-dependent trends in interfacial robustness.

As illustrated in Figure 4A, a progressively increasing normal load was applied via an indenter tip while lateral force and depth were monitored to assess the film's resistance to delamination. The lateral force–displacement profiles (Figure 4B) revealed a clear thickness-dependent response. Generally, the lateral force showed a significant drop, followed by a gradual recovery when delamination of a film or brittle fracture of a substrate occurs. The ZnF₂(45)/Si sample (~30 nm)

exhibited no significant drop, signifying superior interfacial adhesion. In comparison, ZnF₂(90)/Si (~60 nm) and ZnF₂(135)/Si (~90 nm) showed significant drops in measured lateral force, indicating diminished mechanical robustness with increasing thickness.

These observations were corroborated by postscratch SEM analysis. The ZnF₂(45)/Si surface (Figure 4C) remained largely intact with no delamination, demonstrating strong mechanical cohesion. In contrast, the ZnF₂(90)/Si and ZnF₂(135)/Si samples exhibited early signs of delamination

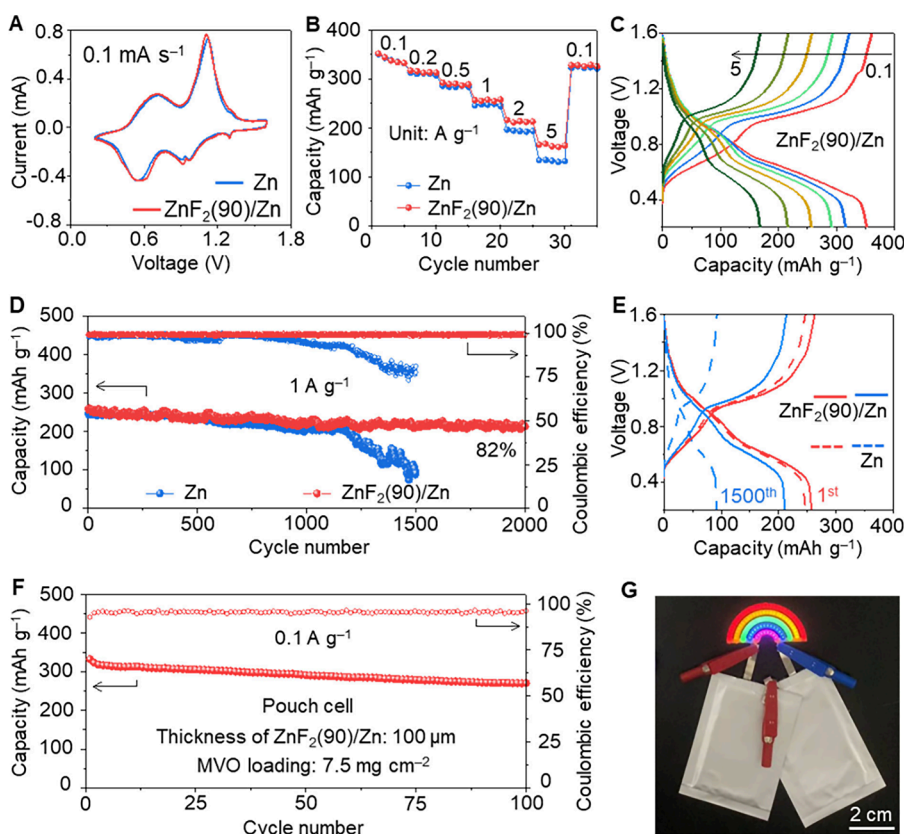


Figure 5. Electrochemical performance of Zn and $\text{ZnF}_2(90)/\text{Zn}$ anodes in full-cell and pouch-cell configurations. (A, B) CV profiles (A) and rate capabilities (B) of full cells employing bare Zn and $\text{ZnF}_2(90)/\text{Zn}$ anodes paired with MVO cathodes. (C) GCD curves of the $\text{ZnF}_2(90)/\text{Zn}$ -based full cell at various current densities, demonstrating stable voltage profiles and high capacity retention. (D) Long-term cycling performance of full cells with Zn and $\text{ZnF}_2(90)/\text{Zn}$ anodes at 1.0 A g^{-1} , showing superior durability for the ZnF_2 -protected electrode. (E) GCD curves recorded at the first and 1500th cycle of the $\text{ZnF}_2(90)/\text{Zn}$ full cell. (F) Cycling performance of a pouch cell incorporating the $\text{ZnF}_2(90)/\text{Zn}$ anode at 0.1 A g^{-1} . (G) Optical image of $\text{ZnF}_2(90)/\text{Zn}$ -based pouch cells successfully powering 3 V LEDs, demonstrating practical applicability.

and more extensive failure patterns (Figure 4D,E), suggesting a decline in interfacial toughness at higher thicknesses. Although the relationship between film thickness and adhesion strength is highly case-specific, the stronger adhesion observed for thinner ZnF_2 films in our study results from a dominance of interfacial (adhesive) bonding relative to cohesive failure within the film. Thinner coatings are more effectively governed by ZnF_2 -substrate interactions and are less likely to accumulate sufficient internal stress to delaminate. By contrast, thicker films are prone to stress build-up, leading to increased probability of cohesive cracking or delamination.³⁵ It is believed that while thinner ZnF_2 coatings may facilitate Zn^{2+} diffusion, excessively strong interfacial adhesion could impede the deposition and dissolution of Zn beneath these layers. In contrast, thicker ZnF_2 coatings may offer more moderate adhesion strength but could suffer from mechanical fragility and hinder Zn^{2+} transport. The observed behavior highlighted a critical trade-off between electrochemical functionality and interfacial adhesion. In this context, the $\sim 90 \text{ nm}$ ZnF_2 layer employed in this work appears to achieve an optimal balance between thickness and adhesion strength (Figure 4F). The layer is thin enough to provide effective ionic modulation and surface protection, yet sufficiently adherent to preserve interfacial contact throughout extended cycling. These findings underscore the importance of simultaneously engineering both electrochemical and mechanical stability, factors that have

often been overlooked or insufficiently investigated, for durable artificial SEI design in Zn-ion batteries. Based on our findings, we believe that systematic evaluation of coating–substrate adhesion will be essential for further optimizing interfacial engineering strategies, which have already proven effective in stabilizing Zn-metal anodes.^{36–39}

FULL-BATTERY CELL PERFORMANCE

To highlight the superiority of the $\text{ZnF}_2(90)/\text{Zn}$, we compared the electrochemical performance of the full cells using either $\text{ZnF}_2(90)/\text{Zn}$ or Zn as the anode and MVO (Figure S11) as the cathode. The cyclic voltammetry (CV) curves display two main cathodic/anodic peak couples at 0.93/1.12 V and 0.57/0.71 V, corresponding to the sequential redox processes of $\text{V}^{5+} \leftrightarrow \text{V}^{4+}$ and $\text{V}^{4+} \leftrightarrow \text{V}^{3+}$ associated with Zn^{2+} insertion and extraction.⁴⁰ Notably, the cell using $\text{ZnF}_2(90)/\text{Zn}$ exhibited redox peaks with higher intensity and lower polarization compared to those of the cell using Zn (Figure 5A), indicating enhanced redox kinetics of the $\text{ZnF}_2(90)/\text{Zn}$ anode.⁴¹ The cell using $\text{ZnF}_2(90)/\text{Zn}$ also demonstrated superior rate capability (Figure 5B). Specifically, at a low current density of 0.1 A g^{-1} , both cells using $\text{ZnF}_2(90)/\text{Zn}$ and Zn delivered similar capacities of 352 mAh g^{-1} . However, at a high current density of 5 A g^{-1} , the capacity of the cell using $\text{ZnF}_2(90)/\text{Zn}$ was 166 mAh g^{-1} , outperforming the cell using Zn (134 mAh g^{-1}). Furthermore, all galvanostatic charge–discharge (GCD)

profiles of the cell using $\text{ZnF}_2(90)/\text{Zn}$ at various current densities exhibited relatively similar shapes (Figure 5C). These results demonstrated that the $\text{ZnF}_2(90)/\text{Zn}$ anode endowed ZIBs with fast charging capability. The excellent stability of the $\text{ZnF}_2(90)/\text{Zn}$ anode was evidently demonstrated in the long term cycling test. The cell using $\text{ZnF}_2(90)/\text{Zn}$ exhibited excellent capacity retention of 82% after 2000 cycles at 1 A g^{-1} while maintaining consistent GCD profiles (Figure 5D,E). In contrast, the capacity of the cell using Zn quickly reduced after 1200 cycles, which may be caused by dendritic growth on the bare Zn anode (Figure S12).⁴⁰ The $\text{ZnF}_2(90)/\text{Zn}$ could still work well when paired with common hydrated vanadium oxide (VOH) (Figure S13). To assess the practical application potential of the $\text{ZnF}_2(90)/\text{Zn}$ anode, pouch cells with high mass loading of MVO (7.5 mg cm^{-2}) were assembled. Their open-circuit voltage was recorded to be 1.342 V (Figure S14). Encouragingly, the pouch cell using $\text{ZnF}_2(90)/\text{Zn}$ displayed high cycling stability for 100 cycles, retaining 80.8% of its initial high capacity (Figure 5F). The high performance could be well preserved even when the thickness is reduced to $50 \mu\text{m}$ (Figure S15). Two pouch cells connected in series successfully powered 3 V light-emitting diodes (LEDs) (Figure 5G), demonstrating the practical viability of the $\text{ZnF}_2(90)/\text{Zn}$ anode.

In summary, we developed a high-performance zinc–metal anode by coating it with an ultrathin ZnF_2 layer via atomic layer deposition. Beyond demonstrating excellent electrochemical metrics—including prolonged symmetric-cell life-spans and high-rate full-cell stability, we introduced nano-scratch-based mechanical testing to evaluate the interfacial toughness of the ZnF_2 layer. Our results indicated that the 90 nm ZnF_2 layer offers an optimal balance: sufficiently thin to suppress dendrite formation and protect against side reactions, yet thick enough to maintain robust mechanical adhesion. Importantly, we showed that increasing film thickness beyond this point leads to a loss of interfacial integrity under stress, as evidenced by lower critical loads and delamination in SEM analysis. This work highlights the critical role of interfacial mechanics in stabilizing metal anodes and sets a new direction for designing artificial SEIs that are not only electrochemically functional but also mechanically resilient. We hope that the methodology and insights presented herein can be broadly extended to other metal-anode systems in next-generation rechargeable batteries.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenergylett.5c02565>.

Experimental details; Supporting figures showing SEM images, TEM images, DFT data, cycling performance; Supporting table (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Viet Phuong Nguyen – Department of Nanomechanics, Korea Institute of Machinery and Materials (KIMM), Daejeon 34103, Republic of Korea; orcid.org/0000-0002-7727-2499; Email: phuongnv9@ust.ac.kr

Seung-Mo Lee – Department of Nanomechanics, Korea Institute of Machinery and Materials (KIMM), Daejeon 34103, Republic of Korea; Nanomechatronics, University of

Science and Technology (UST), Daejeon 34113, Republic of Korea; orcid.org/0000-0003-0732-4473;

Email: sm.lee@kimm.re.kr

Eun-cha Jeon – School of Materials Science and Engineering, University of Ulsan, Ulsan 44776, Republic of Korea; Email: jeonec@ulsan.ac.kr

Authors

Minji Park – School of Materials Science and Engineering, University of Ulsan, Ulsan 44776, Republic of Korea

Young-Woon Byeon – Advanced Analysis and Data Center, Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea

Suim Lim – Energy AI and Computational Science Laboratory, Korea Institute of Energy Research (KIER), Daejeon 34129, Republic of Korea; Department of Mechanical Engineering, Sogang University, Seoul 04107, Republic of Korea

Kanghoon Yim – Energy AI and Computational Science Laboratory, Korea Institute of Energy Research (KIER), Daejeon 34129, Republic of Korea

Minsub Oh – Department of Nanomechanics, Korea Institute of Machinery and Materials (KIMM), Daejeon 34103, Republic of Korea

Seungmin Hyun – Department of Nanomechanics, Korea Institute of Machinery and Materials (KIMM), Daejeon 34103, Republic of Korea; Nanomechatronics, University of Science and Technology (UST), Daejeon 34113, Republic of Korea; orcid.org/0000-0002-7286-9026

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acsenergylett.5c02565>

Notes

The authors declare no competing financial interest.

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